

```
# Experiment 1: Fine-tuning BERT with default parameters
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===== Epoch 1/3 =====

Train loss: 0.3225, Train accuracy: 0.8566

Evaluating...

Test loss: 0.2637, Test accuracy: 0.8863

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.92      | 0.85   | 0.88     | 4951    |
| positive     | 0.86      | 0.92   | 0.89     | 5049    |
| accuracy     |           |        | 0.89     | 10000   |
| macro avg    | 0.89      | 0.89   | 0.89     | 10000   |
| weighted avg | 0.89      | 0.89   | 0.89     | 10000   |

===== Epoch 2/3 =====

Train loss: 0.1911, Train accuracy: 0.9260

Evaluating...

Test loss: 0.2812, Test accuracy: 0.8935

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.91      | 0.88   | 0.89     | 4951    |
| positive     | 0.88      | 0.91   | 0.90     | 5049    |
| accuracy     |           |        | 0.89     | 10000   |
| macro avg    | 0.89      | 0.89   | 0.89     | 10000   |
| weighted avg | 0.89      | 0.89   | 0.89     | 10000   |

===== Epoch 3/3 =====

Train loss: 0.1076, Train accuracy: 0.9620

Evaluating...

Test loss: 0.3131, Test accuracy: 0.8883

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.89      | 0.89   | 0.89     | 4951    |
| positive     | 0.89      | 0.89   | 0.89     | 5049    |
| accuracy     |           |        | 0.89     | 10000   |
| macro avg    | 0.89      | 0.89   | 0.89     | 10000   |
| weighted avg | 0.89      | 0.89   | 0.89     | 10000   |

```
# Experiment 2: Using different amounts of data for training and testing
```

==== Epoch 1/3 =====

Train loss: 0.5995, Train accuracy: 0.6650

Evaluating...

Test loss: 0.5339, Test accuracy: 0.7700

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.79      | 0.71   | 0.75     | 48      |
| positive     | 0.75      | 0.83   | 0.79     | 52      |
| accuracy     |           | 0.77   |          | 100     |
| macro avg    | 0.77      | 0.77   | 0.77     | 100     |
| weighted avg | 0.77      | 0.77   | 0.77     | 100     |

==== Epoch 2/3 =====

Train loss: 0.3421, Train accuracy: 0.8675

Evaluating...

Test loss: 0.4597, Test accuracy: 0.8000

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.80      | 0.77   | 0.79     | 48      |
| positive     | 0.80      | 0.83   | 0.81     | 52      |
| accuracy     |           | 0.80   |          | 100     |
| macro avg    | 0.80      | 0.80   | 0.80     | 100     |
| weighted avg | 0.80      | 0.80   | 0.80     | 100     |

==== Epoch 3/3 =====

Train loss: 0.1401, Train accuracy: 0.9500

Evaluating...

Test loss: 0.9239, Test accuracy: 0.6800

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.60      | 1.00   | 0.75     | 48      |
| positive     | 1.00      | 0.38   | 0.56     | 52      |
| accuracy     |           | 0.68   |          | 100     |
| macro avg    | 0.80      | 0.69   | 0.65     | 100     |
| weighted avg | 0.81      | 0.68   | 0.65     | 100     |

### # Experiment 3: Freezing BERT layers

==== Epoch 1/3 =====

Train loss: 0.6723, Train accuracy: 0.5869

Evaluating...

Test loss: 0.6433, Test accuracy: 0.6634

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.69      | 0.57   | 0.63     | 4951    |
| positive     | 0.64      | 0.75   | 0.69     | 5049    |
| accuracy     |           |        | 0.66     | 10000   |
| macro avg    | 0.67      | 0.66   | 0.66     | 10000   |
| weighted avg | 0.67      | 0.66   | 0.66     | 10000   |

==== Epoch 2/3 =====

Train loss: 0.6381, Train accuracy: 0.6527

Evaluating...

Test loss: 0.6169, Test accuracy: 0.6916

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.72      | 0.61   | 0.66     | 4951    |
| positive     | 0.67      | 0.77   | 0.72     | 5049    |
| accuracy     |           |        | 0.69     | 10000   |
| macro avg    | 0.70      | 0.69   | 0.69     | 10000   |
| weighted avg | 0.70      | 0.69   | 0.69     | 10000   |

==== Epoch 3/3 =====

Train loss: 0.6200, Train accuracy: 0.6758

Evaluating...

Test loss: 0.5999, Test accuracy: 0.7000

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.74      | 0.60   | 0.67     | 4951    |
| positive     | 0.67      | 0.79   | 0.73     | 5049    |
| accuracy     |           |        | 0.70     | 10000   |
| macro avg    | 0.71      | 0.70   | 0.70     | 10000   |
| weighted avg | 0.71      | 0.70   | 0.70     | 10000   |

#### # Experiment 4: Varying learning rates

==== Epoch 1/3 =====

Train loss: 0.3110, Train accuracy: 0.8666

Evaluating...

Test loss: 0.3185, Test accuracy: 0.8650

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.96      | 0.76   | 0.85     | 4951    |
| positive     | 0.81      | 0.97   | 0.88     | 5049    |
| accuracy     |           |        | 0.86     | 10000   |
| macro avg    | 0.88      | 0.86   | 0.86     | 10000   |
| weighted avg | 0.88      | 0.86   | 0.86     | 10000   |

==== Epoch 2/3 =====

Train loss: 0.1838, Train accuracy: 0.9283

Evaluating...

Test loss: 0.2464, Test accuracy: 0.9013

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.90      | 0.90   | 0.90     | 4951    |
| positive     | 0.90      | 0.90   | 0.90     | 5049    |
| accuracy     |           |        | 0.90     | 10000   |
| macro avg    | 0.90      | 0.90   | 0.90     | 10000   |
| weighted avg | 0.90      | 0.90   | 0.90     | 10000   |

==== Epoch 3/3 =====

Train loss: 0.0917, Train accuracy: 0.9672

Evaluating...

Test loss: 0.3321, Test accuracy: 0.8911

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.91      | 0.86   | 0.89     | 4951    |
| positive     | 0.87      | 0.92   | 0.89     | 5049    |
| accuracy     |           |        | 0.89     | 10000   |
| macro avg    | 0.89      | 0.89   | 0.89     | 10000   |
| weighted avg | 0.89      | 0.89   | 0.89     | 10000   |

## # Experiment 5: Varying batch sizes

==== Epoch 1/3 =====

Train loss: 0.3126, Train accuracy: 0.8633

Evaluating...

Test loss: 0.2559, Test accuracy: 0.8927

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.90      | 0.89   | 0.89     | 4951    |
| positive     | 0.89      | 0.90   | 0.89     | 5049    |
| accuracy     |           |        | 0.89     | 10000   |
| macro avg    | 0.89      | 0.89   | 0.89     | 10000   |
| weighted avg | 0.89      | 0.89   | 0.89     | 10000   |

==== Epoch 2/3 =====

Train loss: 0.1798, Train accuracy: 0.9302

Evaluating...

Test loss: 0.2577, Test accuracy: 0.8960

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.89      | 0.91   | 0.90     | 4951    |
| positive     | 0.91      | 0.89   | 0.90     | 5049    |
| accuracy     |           |        | 0.90     | 10000   |
| macro avg    | 0.90      | 0.90   | 0.90     | 10000   |
| weighted avg | 0.90      | 0.90   | 0.90     | 10000   |

==== Epoch 3/3 =====

Train loss: 0.0882, Train accuracy: 0.9689

Evaluating...

Test loss: 0.3140, Test accuracy: 0.8925

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.87      | 0.91   | 0.89     | 4951    |
| positive     | 0.91      | 0.87   | 0.89     | 5049    |
| accuracy     |           |        | 0.89     | 10000   |
| macro avg    | 0.89      | 0.89   | 0.89     | 10000   |
| weighted avg | 0.89      | 0.89   | 0.89     | 10000   |

## # Experiment 6: Varying number of epochs

==== Epoch 1/6 =====

Train loss: 0.3207, Train accuracy: 0.8594

Evaluating...

Test loss: 0.2677, Test accuracy: 0.8868

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.86      | 0.91   | 0.89     | 4951    |
| positive     | 0.91      | 0.86   | 0.88     | 5049    |
| accuracy     |           |        | 0.89     | 10000   |
| macro avg    | 0.89      | 0.89   | 0.89     | 10000   |
| weighted avg | 0.89      | 0.89   | 0.89     | 10000   |

==== Epoch 2/6 =====

Train loss: 0.1863, Train accuracy: 0.9282

Evaluating...

Test loss: 0.2957, Test accuracy: 0.8905

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.90      | 0.88   | 0.89     | 4951    |
| positive     | 0.89      | 0.90   | 0.89     | 5049    |
| accuracy     |           |        | 0.89     | 10000   |
| macro avg    | 0.89      | 0.89   | 0.89     | 10000   |
| weighted avg | 0.89      | 0.89   | 0.89     | 10000   |

==== Epoch 3/6 =====

Train loss: 0.0991, Train accuracy: 0.9641

Evaluating...

Test loss: 0.3311, Test accuracy: 0.8873

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.88      | 0.90   | 0.89     | 4951    |
| positive     | 0.90      | 0.88   | 0.89     | 5049    |
| accuracy     |           |        | 0.89     | 10000   |
| macro avg    | 0.89      | 0.89   | 0.89     | 10000   |
| weighted avg | 0.89      | 0.89   | 0.89     | 10000   |

==== Epoch 4/6 =====

Train loss: 0.0636, Train accuracy: 0.9789

Evaluating...

Test loss: 0.3806, Test accuracy: 0.8888

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.92      | 0.85   | 0.88     | 4951    |
| positive     | 0.86      | 0.93   | 0.89     | 5049    |
| accuracy     |           |        | 0.89     | 10000   |
| macro avg    | 0.89      | 0.89   | 0.89     | 10000   |
| weighted avg | 0.89      | 0.89   | 0.89     | 10000   |

==== Epoch 5/6 =====

Train loss: 0.0460, Train accuracy: 0.9849

Evaluating...

Test loss: 0.4778, Test accuracy: 0.8851

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.90      | 0.87   | 0.88     | 4951    |
| positive     | 0.87      | 0.90   | 0.89     | 5049    |
| accuracy     |           |        | 0.89     | 10000   |
| macro avg    | 0.89      | 0.88   | 0.89     | 10000   |
| weighted avg | 0.89      | 0.89   | 0.89     | 10000   |

==== Epoch 6/6 =====

Train loss: 0.0411, Train accuracy: 0.9867

Evaluating...

Test loss: 0.4015, Test accuracy: 0.8893

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.88      | 0.90   | 0.89     | 4951    |
| positive     | 0.90      | 0.88   | 0.89     | 5049    |
| accuracy     |           |        | 0.89     | 10000   |
| macro avg    | 0.89      | 0.89   | 0.89     | 10000   |
| weighted avg | 0.89      | 0.89   | 0.89     | 10000   |

## # Experiment 7: Freezing partial BERT layers

==== Epoch 1/3 =====

Test loss: 0.2607, Test accuracy: 0.8897

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.89      | 0.88   | 0.89     | 4951    |
| positive     | 0.89      | 0.90   | 0.89     | 5049    |
| accuracy     |           |        | 0.89     | 10000   |
| macro avg    | 0.89      | 0.89   | 0.89     | 10000   |
| weighted avg | 0.89      | 0.89   | 0.89     | 10000   |

==== Epoch 2/3 =====

Train loss: 0.1891, Train accuracy: 0.9249

Evaluating...

Test loss: 0.2978, Test accuracy: 0.8896

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.86      | 0.92   | 0.89     | 4951    |
| positive     | 0.92      | 0.86   | 0.89     | 5049    |
| accuracy     |           |        | 0.89     | 10000   |
| macro avg    | 0.89      | 0.89   | 0.89     | 10000   |
| weighted avg | 0.89      | 0.89   | 0.89     | 10000   |

==== Epoch 3/3 =====

Train loss: 0.1039, Train accuracy: 0.9624

Evaluating...

Test loss: 0.3047, Test accuracy: 0.8903

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.93      | 0.84   | 0.88     | 4951    |
| positive     | 0.86      | 0.94   | 0.90     | 5049    |
| accuracy     |           |        | 0.89     | 10000   |
| macro avg    | 0.89      | 0.89   | 0.89     | 10000   |
| weighted avg | 0.89      | 0.89   | 0.89     | 10000   |

## # Experiment 8: Varing the max length

==== Epoch 1/3 =====

Train loss: 0.2667, Train accuracy: 0.8909

Evaluating...

Test loss: 0.2071, Test accuracy: 0.9203

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.91      | 0.94   | 0.92     | 4951    |
| positive     | 0.94      | 0.90   | 0.92     | 5049    |
| accuracy     |           |        | 0.92     | 10000   |
| macro avg    | 0.92      | 0.92   | 0.92     | 10000   |
| weighted avg | 0.92      | 0.92   | 0.92     | 10000   |

==== Epoch 2/3 =====

Train loss: 0.1489, Train accuracy: 0.9445

Evaluating...

Test loss: 0.2453, Test accuracy: 0.9173

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.89      | 0.95   | 0.92     | 4951    |
| positive     | 0.94      | 0.89   | 0.92     | 5049    |
| accuracy     |           |        | 0.92     | 10000   |
| macro avg    | 0.92      | 0.92   | 0.92     | 10000   |
| weighted avg | 0.92      | 0.92   | 0.92     | 10000   |

==== Epoch 3/3 =====

Train loss: 0.0819, Train accuracy: 0.9714

Evaluating...

Test loss: 0.2575, Test accuracy: 0.9208

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.91      | 0.93   | 0.92     | 4951    |
| positive     | 0.93      | 0.91   | 0.92     | 5049    |
| accuracy     |           |        | 0.92     | 10000   |
| macro avg    | 0.92      | 0.92   | 0.92     | 10000   |
| weighted avg | 0.92      | 0.92   | 0.92     | 10000   |

## # Experiment 9: Varing dropout rate

===== Epoch 1/3 =====

Train loss: 0.3613, Train accuracy: 0.8418

Evaluating...

Test loss: 0.3244, Test accuracy: 0.8642

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.94      | 0.78   | 0.85     | 4951    |
| positive     | 0.81      | 0.95   | 0.88     | 5049    |
| accuracy     |           |        | 0.86     | 10000   |
| macro avg    | 0.88      | 0.86   | 0.86     | 10000   |
| weighted avg | 0.88      | 0.86   | 0.86     | 10000   |

===== Epoch 2/3 =====

Train loss: 0.2794, Train accuracy: 0.8827

Evaluating...

Test loss: 0.2960, Test accuracy: 0.8774

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.87      | 0.89   | 0.88     | 4951    |
| positive     | 0.89      | 0.87   | 0.88     | 5049    |
| accuracy     |           |        | 0.88     | 10000   |
| macro avg    | 0.88      | 0.88   | 0.88     | 10000   |
| weighted avg | 0.88      | 0.88   | 0.88     | 10000   |

===== Epoch 3/3 =====

Train loss: 0.2302, Train accuracy: 0.9073

Evaluating...

Test loss: 0.2991, Test accuracy: 0.8908

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.91      | 0.87   | 0.89     | 4951    |
| positive     | 0.87      | 0.92   | 0.89     | 5049    |
| accuracy     |           |        | 0.89     | 10000   |
| macro avg    | 0.89      | 0.89   | 0.89     | 10000   |
| weighted avg | 0.89      | 0.89   | 0.89     | 10000   |

```
# Experiment 10: Using the output embeddings of other tokens instead of that of the
class token
```

==== Epoch 1/3 =====

Train loss: 0.3136, Train accuracy: 0.8639

Evaluating...

Test loss: 0.2784, Test accuracy: 0.8757

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.84      | 0.93   | 0.88     | 4951    |
| positive     | 0.93      | 0.82   | 0.87     | 5049    |
| accuracy     |           |        | 0.88     | 10000   |
| macro avg    | 0.88      | 0.88   | 0.88     | 10000   |
| weighted avg | 0.88      | 0.88   | 0.88     | 10000   |

==== Epoch 2/3 =====

Train loss: 0.1846, Train accuracy: 0.9278

Evaluating...

Test loss: 0.2647, Test accuracy: 0.8961

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.89      | 0.90   | 0.90     | 4951    |
| positive     | 0.90      | 0.89   | 0.90     | 5049    |
| accuracy     |           |        | 0.90     | 10000   |
| macro avg    | 0.90      | 0.90   | 0.90     | 10000   |
| weighted avg | 0.90      | 0.90   | 0.90     | 10000   |

==== Epoch 3/3 =====

Train loss: 0.0934, Train accuracy: 0.9654

Evaluating...

Test loss: 0.3206, Test accuracy: 0.8889

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| negative     | 0.91      | 0.86   | 0.88     | 4951    |
| positive     | 0.87      | 0.91   | 0.89     | 5049    |
| accuracy     |           |        | 0.89     | 10000   |
| macro avg    | 0.89      | 0.89   | 0.89     | 10000   |
| weighted avg | 0.89      | 0.89   | 0.89     | 10000   |

